RYAN BARRY

Game Programmer · Gameplay, AI & Engine Systems

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I want to get better at making games every day, and most of my best work was done by choice, outside the curriculum: a custom engine for my award-winning game **Pale Roots**, a research paper on hybrid ECS traffic, a sandbox to teach myself optimisation, **Tallow** with an AI-integrated workflow, and my own portfolio site. All of this alongside a **First Class Honours** average and tech-lead duties on a four-person team.

TECHNICAL SKILLS

Languages	C#, C++, SQL, Kotlin, XAML
Engines	Unity, Unreal Engine, MonoGame, WPF, .NET
Architecture	Data-oriented design, event-driven design, State, Strategy, Factory, Chain of Responsibility, MVVM, dependency inversion
Performance	Unity Profiler, ECS with the Burst compiler and Jobs, simulation level of detail, object pooling, garbage-free per-frame code, throttled AI updates, async asset loading, GPU instancing, occlusion culling
Maths & physics	Pure-pursuit steering from wheelbase geometry, cornering speed from turn radius and grip, braking-distance and car-following kinematics, throttle and brake feedback control, Bézier lane curves, drift physics, custom 2D collision and forces
Testing & tools	Unity Test Framework (NUnit), allocation tests, command-line test runs, Git & GitHub, Visual Studio, VS Code, Trello
AI tools	Claude Code with the Unity MCP server, custom subagents and test-checked changes, NotebookLM for study material

PROJECTS

Pale Roots — EA Award · Games Fleadh 2026

Solo · C#, MonoGame · Dec 2025 – Mar 2026

Sci-fi roguelike on my own engine, where the enemies you kill make you stronger.

- Engineered the engine layer myself: game loop, camera, collision, animation, depth-sorted rendering and a game-state stack.
- Award-winning AI: allies and enemies share one state machine (Charge, Wander, Chase, Combat, Hurt) behind an *IAIState* interface. A targeting system re-scans every 500 ms and caps each target at two attackers, splitting 15+ unit battles into readable duels.
- Six spells extend one abstract *Spell* class with cooldowns and timed effects that apply and then cleanly revert, offered through level-up cards that carry their effect as an *Action* delegate.

Play on itch.io · Source · Write-up

Hybrid LOD Traffic Simulation — Research paper

Solo research · Unity DOTS/ECS, Burst · 2026

Open-world traffic that switches each car between a full GameObject and lightweight ECS data.

- Within 150 m of the camera a car is a full GameObject, and past 200 m it becomes an ECS entity moved kinematically by a Burst-compiled job. Shared integer node IDs carry its position, destination and remaining A* path across.
- 150 pre-warmed pooled cars mean a hand-off never instantiates anything during play.
- Profiled 100 cars three ways: all GameObjects (rigidbodies, raycasts, Update loops) gave GC spikes and high frame times, pure ECS removed GC, and the hybrid kept frame times stable with full detail near the player.

Play in browser · Read the paper · Write-up

PROJECTS (CONTINUED)

Split Shift — Tech lead

Team of 4 · Unity, C# · Sept 2026 – present

Open-world street racing game, aiming for Games Fleadh 2027. I engineered the AI, traffic, police and city generation.

- Modular AI driver: a sensor module and a stateless decision module pass structs through a central brain, which drives any car through one *IVehicleInput* interface. Driver profiles set patience and style.
- Police pursuit: a reported crime sends the three nearest units, re-routed to the suspect every second. Cars steer with pure-pursuit path tracking, with traffic, racing and police behaviour as swappable strategies.
- As tech lead, split the project into 12 assemblies with one-way dependencies, run sprint planning and reviews, and keep 59 automated tests passing, including allocation tests on the AI's per-frame code.

Play in browser · Write-up

Tallow — AI-integrated workflow

Solo · Unity, C# · Oct 2026

Candlelit horror card-duel roguelike. Built with an AI-integrated workflow, to show I can work in modern development pipelines as well as craft systems by hand.

- I set the design, architecture and rules. Game logic is plain C# in its own assembly that cannot reference Unity, and raises events the view replays as animation, so every rule is tested without a scene (113 automated tests).
- Balanced by having a bot play 400 runs on fixed random seeds, then retuning the boss from its win rate.

Play in browser · Write-up

Other projects

- **Dying Fire** (C#, WPF, SQLite): horror visual novel built on Strategy and Chain of Responsibility. Source
- **Paranormal Investigation** (Unity): seven-state ghost AI with allocation-free detection and event channels. Play
- **Engine Architecture & Optimisation** (Unity): 500 pooled enemies at 84 FPS. Play

SOFT SKILLS

Leadership	Tech lead on a four-person game team. Lead a bar team and train new staff.
Teamwork	Six years in hospitality teams, taking direction across eight venues.
Communication	Clear technical writing (research paper, portfolio) and daily customer contact.
Under pressure	Run busy service nights alongside full-time study and a First Class average.

EDUCATION

BSc Computer Games Development, ATU Sligo

Sept 2023 – 2027 (expected)

Current average: **First Class Honours (1:1)**

3D Game Programming	**95**	Mathematics 4	**76**
Database Technology	**88**	Data Structures & Algorithms	**72**
Software Quality & Testing	**81**	Object-Oriented Development	**71**
2D Game Programming	**78**		

This year: Advanced Games Programming, Rich App Development, Mobile App Development and C++.

Leaving Certificate, Ardgillan Community College, Dublin — 520 points

EXPERIENCE

Manager & Head Bartender

Lillies Cocktail Bar & Tannery Lane, Sligo · 2024 – present

- Lead the floor team on busy nights, write the rota, manage stock and train new staff.

Bartender

Bourbon, Kennedy's, Lola's & The Belfry, Sligo · 2023 – 2024

- Joined a new team and worked a different venue each night, under several managers and supervisors.

Supervisor

Blueberry Hill Café, Dublin · 2021 – 2023

- Ran the bar and stepped into the kitchen whenever service needed it.

Head Bartender

Lala Café (hotel restaurant), Dublin · 2020 – 2021

- Promoted to head bartender at 17, running the bar during busy hotel service.